

Aleksandar Anžel

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Aleksandar Anžel

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Born 06.08.1995.

WORK EXPERIENCE

March 2025 – present

Deputy Group Leader

Robert Koch Institute, Center for Artificial Intelligence in Public Health Research (ZKI-PH 5: Visualization), Berlin

- Assisted in managing research projects, supervising group members, and supporting the recruitment of new team members. Supported strategic planning and decision-making processes. Coordinated collaborations and administrative tasks within the group. Collaborated with other departments and institutions to establish interdisciplinary linkages and foster cross-disciplinary collaborations. Partnered with leading organizations such as the World Health Organization (WHO), the German Ministry of Health (BMG), and similar institutions on numerous projects, enhancing the group's impact and reach. Authored and contributed to several internal and external grant proposals, successfully securing funding to support ongoing and new research initiatives.

July 2024 – October 2026

Visiting Research Scientist

Das Zentralinstitut für die kassenärztliche Versorgung in der Bundesrepublik Deutschland (Zi), Berlin

- Provided guidance, mentorship, and technical expertise on a cutting-edge AI surveillance system project that leveraged medical data from across Germany. Demonstrated excellent communication and leadership skills to drive the project forward and translate complex technical concepts into actionable insights. Leveraged management experience to oversee all aspects of the initiative, from risk mitigation to resource allocation. Documented findings, published research, and presented results.

October 2023 – present

Postdoctoral Researcher

Robert Koch Institute, Center for Artificial Intelligence in Public Health Research (ZKI-PH 5: Visualization), Berlin

- Contributed to cutting-edge research and development efforts at the intersection of artificial intelligence (AI), machine learning, visualization, and public health. Collaborated with a multidisciplinary team of researchers to design and implement novel techniques for visualizing and explaining AI models in public health applications. Participated in classified AI projects aimed at enhancing the country's preparedness for large-scale public health emergencies. Developed innovative AI-driven solutions to analyze complex biological data sets, identify potential biomarkers for early disease detection, and model the spread of infectious agents at a national scale. Conducted research and published findings in reputable scientific journals or presented at relevant international conferences.

July 2022 – February 2023

Technical Lead

eMedicals Healthtech GmbH (currently Oxalis DeepBio GmbH), Frankfurt am Main

- Mentored and led two teams of 8 people to increase productivity and maximize product quality. Helped with discovering and hiring new team members. Assessed used technologies and advised on SOTA the team could use. Determined potential risks and proposed ways to mitigate them, thus providing GDPR compliant and secure internal platform. Asserted data compliance and ensured its proper adoption. Used Scrum methodology to help the product owner and acted as a Scrum Master. Designed and implemented application-critical parts of the *kidi* platform. Improved the performance of legacy code and decreased loading times by 40%. Reviewed the code of other team members and promoted collaboration and engagement between two teams. Provided architectural and design directions for the company's products and reinforced the company's goals. Guided the development according to multiple SaMD standards and conducted internal audits.

October 2021 – July 2022

Data Scientist

eMedicals Healthtech GmbH (currently Oxalis DeepBio GmbH), Frankfurt am Main

- Used analytical, statistical, and programming skills to collect, analyze, and interpret large medical and biological data sets. Streamlined algorithm development, machine learning workflows (*i.e.,* NLP and computer vision), and statistical techniques to produce solutions to problems quickly. Initiated automated testing efforts that reduced post production defects by 30%. Closely worked with university partners (*i.e.*, Technische Universität Darmstadt) and research organizations (*i.e.*, Fraunhofer-Gesellschaft zur Förderung der angewandten Forschung e.V.) to improve, evaluate, and further develop algorithms relevant to the company. Closely monitored SOTA AI and data science methods and technologies to support the evolution of the company's strategy.

December 2020 – September 2023

Research Associate

Heider Lab, Philipps-Universität Marburg, Marburg

- Developed bioinformatics and ML pipelines for solving various omics and multi-omics problems. Adapted existing and designed and implemented new visualization methods for biological and medical data sets. Organized and facilitated workshops, seminars, and lectures to empower and inspire students to reach their full potential. Mentored multiple undergraduate and graduate students during their studies and while conducting research. Introduced undergraduate and graduate students from various disciplines to good scientific practices and conventions.

MOSLA (*Molekulare Speicher zur Langzeit-Archivierung*), Philipps-Universität Marburg, Marburg

- Designed, implemented, and evaluated automatic workflows for information storage systems based on molecular storage media. Adapted existing and developed new visualization methods and UIs for novel data storage technologies. Systematically reviewed existing DNA storage tools and methods and promoted DNA as a data storage medium. Actively engaged with other domain specialists and supported and steered the research.

EDUCATION

December 2020 – December 2023

Doctoral degree in Computer Science (Dr. rer. nat.)

Philipps-Universität Marburg, Faculty of Mathematics and Computer Science

- **Anžel, A.** (2023, October). *A tale of two approaches: Comparing top-down and bottom-up strategies for analyzing and visualizing high-dimensional data* [Doctoral dissertation, Philipps-Universität Marburg]. Philipps-Universität Marburg. https://doi.org/10.17192/z2023.0533

August 2021

OxML Summer School

Modules: Representation learning and statistical ML, NLP, ML in healthcare

Machine Learning Summer School, University of Oxford

October 2018 – January 2020

Master's degree in Mathematics

Module: Computer Science and Informatics

University of Belgrade, Faculty of Mathematics

- Average grade: 10.00 (out of 10.00)
- **Anžel, A.** (2020, February). *Determining protein n-glycosylation with machine learning methods* [Master's thesis]. Faculty of Mathematics, University of Belgrade [Available at http://elibrary.matf.bg.ac.rs/bitstream/handle/123456789/5013/AnzelAleksandar.pdf?sequence=1]

October 2014 – July 2018

Bachelor's degree in Mathematics

Module: Computer Science and Informatics

University of Belgrade, Faculty of Mathematics

- Average grade: 8.66 (out of 10.00)

SKILLS

Languages

Serbian — Native proficiency

English — Bilingual proficiency

- Cambridge English (acquired in 2012): First (FCE): upper intermediate (B2 in CEFR)

German — Limited working proficiency

- Goethe-Institut e. V., In-house (acquired in 2026): upper intermediate (B2 in CEFR)
- telc gGmbH German (acquired in 2026): intermediate (B1 in CEFR)

Computer Science

Software Development

Python, C, C++, Java, MATLAB, Shell, Haskell, Assembly IA-64, Assembly ARM-32

Machine Learning

Keras, Tensorflow, Scikit-learn, PyTorch

Data Management

SQL, NoSQL, Pandas, Polars

Visualization

Vega-Lite, Altair, Plotly, Matplotlib

Additional skills

Scientific and Cluster Computing, High-Level Visualization, CI/CD, Containerization (Docker), Cloud Development (Microsoft Azure), Agile Development, Scrum, Kanban, Jira, Code Verification and Validation

Document manipulation

LaTeX, Libre Office Suite, Microsoft Office Suite

Soft skills

- Excellent organizational and communication skills utilized to convey complex scientific concepts to diverse audiences
- Outstanding interpersonal skills leveraged to cultivate productive relationships with academic and industry partners
- Thoroughness, with rigorous attention to both detail and quality
- Project management expertise applied in overseeing research endeavors within academic and industry contexts, delivering results with efficiency and excellence
- Strong analytical thinking skills utilized to drive data-driven decision-making in complex scientific projects

PUBLICATIONS

Peer-Reviewed

2026

- Awotoro, E., **Anžel, A.,** Anyabolu, C., Willrich, N., & Hattab, G. (2026). Network-based characterization of latent co-testing patterns in antimicrobial susceptibility testing. *iScience*, *29*(10). https://doi.org/10.1016/j.isci.2026.117421
- Dubey, A., **Anžel, A.,** İlgen, B., & Hattab, G. (2026a). Ubiqvision: Spatial dempster-shafer fusion of xai attributions for reliable deep vision ensembles. *Machine Learning with Applications*, 101000. https://doi.org/https://doi.org/10.1016/j.mlwa.2026.101000
- **Anžel, A.,** Yang, Z., & Hattab, G. (2026). A multi-technique approach for improving summary polar diagrams. *iScience*, *29*(9). https://doi.org/10.1016/j.isci.2026.117044
- **Anžel, A.,** & Hattab, G. (2026). Limitations of interaction techniques in modern radial chart implementations. *Discover Computing*, *29*(1), 500. https://doi.org/10.1007/s10791-026-10416-2
- Schilling, V., Dubey, A., **Anžel, A.,** & Hattab, G. (2026). Peptrix: A framework for explainable peptide analysis through protein language models. *Next Research*, *12*, 102151. https://doi.org/10.1016/j.nexres.2026.102151
- Ferreira, A. P. G., **Anžel, A.,** Marcilio, I., Hughes, H., Elliot, A. J., Kong, J. D., Schranz, M., Ullrich, A., & Hattab, G. (2026). The open syndrome definition as a machine-readable standard for public health: Design and implementation study. *J Med Internet Res*, *28*, e86249. https://doi.org/10.2196/86249
- Wijaya, A. J., **Anžel, A.,** & Hattab, G. (2026). Evaluating ensemble learning approaches for horizontal gene transfer detection. *Scientific Reports*, *16*(1), 16582. https://doi.org/10.1038/s41598-026-53037-x
- Hassan, H. A., **Anžel, A.,** İlgen, B., Luchner, M., Schramowski, P., Blasse, A., Mayer, J. U., Ladewig, K., Hattab, G., & Sprang, M. (2026). A safety-centric perspective on innovation and risk in the use of artificial intelligence in genomics. *Trends in Genetics*. https://doi.org/10.1016/j.tig.2026.04.001

- Schilling, V., **Anžel, A.,** Doellinger, J., Lasch, P., Heider, D., & Hattab, G. (2026). The state of peptide detectability in computational proteomics and guidelines for ai applications. *Computational and Structural Biotechnology Journal*, *35*(1), 0037. https://doi.org/10.34133/csbj.0037
- Gomes Ferreira, A. P., **Anžel, A.,** Ullrich, A., & Hattab, G. (2026). Advocating the potential of artificial intelligence for syndrome discovery in syndromic surveillance systems: A scoping review. *iScience*, *29*(3). https://doi.org/10.1016/j.isci.2026.115103
- Dubey, A., **Anžel, A.,** İlgen, B., & Hattab, G. (2026b). Ubiqtree: Uncertainty quantification in xai with tree ensembles. *Patterns*, *7*(4). https://doi.org/10.1016/j.patter.2025.101454

2025

- Wijaya, A. J., **Anžel, A.,** Richard, H., & Hattab, G. (2025b). Genomic data representations for horizontal gene transfer detection. *NAR Genomics and Bioinformatics*, *7*(4), lqaf165. https://doi.org/10.1093/nargab/lqaf165
- Hattab, G., **Anžel, A.,** Dubey, A., Ezekannagha, C., Yang, Z., & undefinedlgen, B. (2025). Persona adaptable strategies make large language models tractable. *Proceedings of the 2024 8th International Conference on Natural Language Processing and Information Retrieval*, 24–31. https://doi.org/10.1145/3711542.3711581
- Dubey, A., Yang, Z., **Anžel, A.,** & Hattab, G. (2025). Protocol for implementing the nested model for ai design and validation in compliance with ai regulations. *STAR Protocols*, *6*(2), 103771. https://doi.org/10.1016/j.xpro.2025.103771
- Wijaya, A. J., **Anžel, A.,** Richard, H., & Hattab, G. (2025a). Current state and future prospects of horizontal gene transfer detection. *NAR Genomics and Bioinformatics*, *7*(1), lqaf005. https://doi.org/10.1093/nargab/lqaf005

2024

- Weckbecker, M., **Anžel, A.,** Yang, Z., & Hattab, G. (2024). Interpretable molecular encodings and representations for machine learning tasks. *Computational and Structural Biotechnology Journal*, *23*, 2326–2336. https://doi.org/10.1016/j.csbj.2024.05.035
- Yang, Z., Dai, X., Yang, W., Ilgen, B., **Anžel, A.,** & Hattab, G. (2024). Kernel-based learning for safe control of discrete-time unknown systems under incomplete observations. *2024 43rd Chinese Control Conference (CCC)*, 6627–6632. https://doi.org/10.23919/CCC63176.2024.10662288

2023

- **Anžel, A.,** Heider, D., & Hattab, G. (2023). Interactive polar diagrams for model comparison. *Computer Methods and Programs in Biomedicine*, *242*, 107843. https://doi.org/10.1016/j.cmpb.2023.107843
- **Anžel, A.** (2023, October). *A tale of two approaches: Comparing top-down and bottom-up strategies for analyzing and visualizing high-dimensional data* [Doctoral dissertation, Philipps-Universität Marburg]. Philipps-Universität Marburg. https://doi.org/10.17192/z2023.0533
- Hattab, G., **Anžel, A.,** Spänig, S., Neumann, N., & Heider, D. (2023). A parametric approach for molecular encodings using multilevel atomic neighborhoods applied to peptide classification. *NAR Genomics and Bioinformatics*, *5*(1), lqac103. https://doi.org/10.1093/nargab/lqac103

2022

- **Anžel, A.,** Heider, D., & Hattab, G. (2022). Movis: A multi-omics software solution for multi-modal time-series clustering, embedding, and visualizing tasks. *Computational and Structural Biotechnology Journal*, *20*, 1044–1055. https://doi.org/10.1016/j.csbj.2022.02.012

2021

- **Anžel, A.,** Heider, D., & Hattab, G. (2021). The visual story of data storage: From storage properties to user interfaces. *Computational and Structural Biotechnology Journal*, *19*, 4904–4918. https://doi.org/10.1016/j.csbj.2021.08.031

2020

- **Anžel, A.** (2020, February). *Determining protein n-glycosylation with machine learning methods* [Master's thesis]. Faculty of Mathematics, University of Belgrade [Available at http://elibrary.matf.bg.ac.rs/bitstream/handle/123456789/5013/AnzelAleksandar.pdf?sequence=1]

Under Review

2026

- Gomes Ferreira, A. P., **Anžel, A.,** Tavares Veras, P., Pereira Ramos, P. I., Barral-Netto, M., Marcilio, I., & Hattab, G. (2026). *A multi-view similarity network fusion framework for syndrome discovery from aggregated health records*. https://doi.org/10.64898/2026.06.30.26356975
- **Anžel, A.**, Schilling, V., Grevens Carpi, R., Dubey, A., & Hattab, G. (2026). *PeptriX: A PTM-Focused Protein Embedding and Classification Model*.

2025

- **Anžel, A.,** Anyabolu, C., Wimbes, L., Staus, L., Heldian, I. T., Sonnabend, D., Elhadri, K., Becker, S., Klein, F., Schoen, R., Schwarz, M., Welzel, M., Freisleben, B., Heider, D., & Hattab, G. (2026). The unified evaluation app for dna data storage codecs [Available at https://arxiv.org/abs/2608.09673]
- Dubey, A., **Anžel, A.,** & Hattab, G. (2025). *Surrogate interpretable graph for random decision forests*. https://doi.org/10.48550/arXiv.2506.01988

GRANTS AND FUNDING

November 2026 – November 2029

Principal Investigator (PI), "Molecular Embeddings via Low-Distortions Framework for Antimicrobial Resistance (MELD4AMR)", Federal Ministry of Research, Technology and Space (BMFTR), 01DS26014, €300k total funding, Hosted at Robert Koch Institute / Freie Universität Berlin

SOFTWARE PROJECTS

Bioinformatics

- UbiQVision: Uncertainty Quantification in XAI for Image Recognition
- UbiQTree: Uncertainty Quantification in XAI for Tree Ensembles
- PepTriX: A Framework for Explainable Peptide Analysis through Protein Language Models
- iCAN — Interpretable Carbon-based Array of Neighborhoods
- MOVIS: A Multi-Omics Software Solution for Multi-modal Time-Series Clustering, Embedding, and Visualizing Tasks
- CMANGOES — Carbon-based Multi-level AtomicNeiGhborhOod EncodingS
- Determining protein N-glycosylation using machine learning methods
- Modification and analysis of UPGMA algorithm while using different metrics

Computer Science

- OSI: Open Syndrome Initiative
- Enhanced Interactive Summary Polar Diagrams
- Interactive Summary Polar Diagrams for Model Comparison
- Finding Waldo using various Machine Learning methods
- Image modification and correction in Python
- Determining integer variable ranges using Abstract Interpretation in C++ (LLVM, Clang)
- AVL trees in C programming language

SELECTED EVENTS

2026

- *The German Conference on Bioinformatics (GCB2026).* Saarbrücken, Germany. (presenter)
- *Artificial Intelligence in Public Health Research.* Berlin, Germany. (organizer, attendee)

2025

- *Bioinformatics and Computational Biology Conference (BBCC 2025).* Naples, Italy. (presenter)
  - **Anžel, A.** (2025, November). *Enhancing biomedical time series analysis and visualization through interactive summary polar diagrams* [Presented at BBCC2025]. https://doi.org/10.7490/f1000research.1120409.1
- *Artificial Intelligence in Public Health Research.* Berlin, Germany. (presenter)

2024

- *Publication Workshop for Health Professionals: Communicating Insights.* Berlin, Germany. (presenter)
- *Good Scientific Practice: From Data to AI.* Berlin, Germany. (presenter)
- *Serbian Society for Bioinformatics and Computational Biology (BiRBi) Seminar.* Virtual. (presenter)

2023

- *Artificial Intelligence in Public Health Research.* Berlin, Germany. (organizer, attendee)
- *Future Bioinformatics Workshop.* Herborn, Germany. (attendee, co-host)

2022
- *Future Bioinformatics Workshop*. Ebsdorfergrund, Germany. (presenter, co-host)
- *The 1st International Conference on Data Storage in Molecular Media*. Virtual. (attendee, co-host)

2021
- *Symposium on Interdisciplinary Bioinformatics and Biomedical Data Science (IBBMDS)*. Marburg, Germany. (presenter)

TEACHING

2026
- **Guest lecture**, *Advanced Data Visualization for Artificial Intelligence*. Department of Mathematics and Computer Science, Freie Universität Berlin. Berlin, Germany. (presenter)

2022 – 2023
- **Lecture**, *Machine Learning*. Department of Mathematics and Computer Science, Philipps-Universität Marburg. Marburg, Germany. (collaborator, tutor)

2022
- **Guest lecture**, *Molecules as storage media for long-term data storage*. Faculty of Mathematics, University of Belgrade. Belgrade, Serbia. (presenter)

2021 – 2022
- **Group student project (Projektarbeit)**, *DNA Storage Encodings*. Department of Mathematics and Computer Science, Philipps-Universität Marburg. Marburg, Germany. (co-organizer, tutor)

2020 – 2022
- **Lecture**, *Data Visualization*. Department of Mathematics and Computer Science, Philipps-Universität Marburg. Marburg, Germany. (collaborator)
- **Seminar**, *Information Theory Tools for Visual Computing*. Department of Mathematics and Computer Science, Philipps-Universität Marburg. Marburg, Germany. (co-organizer, presenter)

ADVISING & SUPERVISION

April 2025 – present
Ebenezer Awotoro, Ph.D. level, Computer Science (CS). *Antimicrobial resistance surveillance through innovative ML/AI-driven data visualization*. Department of Mathematics and Computer Science, Freie Universität Berlin. Berlin, Germany.

April 2025 – November 2025
Rebecca Grevens Carpi, B.Sc. level, Computer Science (CS). *Predicting Post-translational Modifications with a New Parametric Fingerprinting Method and Cutting-Edge Machine Learning*. Department of Mathematics and Computer Science, Freie Universität Berlin. Berlin, Germany.

September 2024 – February 2026
Akshat Dubey, Ph.D. level, Computer Science (CS).
- Dubey, Akshat. (2026). *Human-in-the-loop and storytelling-based explainable frameworks for the next generation of explainable ensemble methods and artificial intelligence techniques* [Doctoral dissertation, Freie Universität Berlin]. Freie Universität Berlin. https://doi.org/10.17169/REFUBIUM-53170

January 2024 – present
Vincent Schilling, Ph.D. level, Computer Science (CS). *AI-supported proteomics analysis for an effective antimicrobial therapy decision*. Department of Mathematics and Computer Science, Freie Universität Berlin. Berlin, Germany.

October 2023 – present
Andre Jatmiko Wijaya, Ph.D. level, Computer Science (CS). *Monitoring Antimicrobial Resistance (AMR) Reservoirs and the Evolution of Virulence through AI-supported Next Generation Annotation of Horizontal Gene Transfer*. Department of Mathematics and Computer Science, Freie Universität Berlin. Berlin, Germany.

October 2023 – present
Ana Gomes Ferreira, Ph.D. level, Computer Science (CS). *Unsupervised Learning for Surveillance Indicators*. Department of Mathematics and Computer Science, Freie Universität Berlin. Berlin, Germany.

November 2022 – September 2023
Bianca Thiel, M.Sc. level, Computer Science (CS). *Interactive Information-Theoretic Visualization for Plot Types*. Department of Mathematics and Computer Science, Philipps-Universität Marburg. Marburg, Germany.

October 2022 – September 2023 Dilekcan Pamir, B.Sc. level, Computer Science (CS). *Qualitative and Quantitative Visualization of Soft-Tissue Registration Mediated with Isolines*. Department of Mathematics and Computer Science, Philipps-Universität Marburg. Marburg, Germany.

June 2022 – October 2022 Jing Chen, B.Sc. level, Bioinformatics (BI). *Geospatial Visualization of Lake Microbiomes*. Department of Mathematics and Computer Science, Philipps-Universität Marburg. Marburg, Germany.

October 2021 – November 2022 Fabio Rougier, M.Sc. level, Computer Science (CS). *Explainable Machine Learning – Visualization of Random Forests*. Department of Mathematics and Computer Science, Philipps-Universität Marburg. Marburg, Germany.

REVIEWING

2026 – present
- EuroTech Federation, AI Research Summer, RNA: W751283025
- PCI Mathematical and Computational Biology, ISSN: 2803-9963

2025 – present
- PLOS One, ISSN: 1932-6203

2024 – present
- Digital Discovery, ISSN: 2635-098X
- PLOS Computational Biology, ISSN: 1553-7358

2022 – present
- BMC Bioinformatics, ISSN: 1471-2105
- Scientific Reports, ISSN: 2045-2322
- BioData Mining, ISSN: 1756-0381
- Bioinformatics Advances, ISSN: 2635-0041
- Frontiers in Cellular and Infection Microbiology, ISSN: 2235-2988

ADDITIONAL INFORMATION

Scientific Committees and Societies
- Professional member, German Alliance for Global Health Research (GLOHRA). Secretariat c/o Charité Universitätsmedizin Berlin Charitéplatz 1, 10117 Berlin, Germany.
- Member, Serbian Society for Bioinformatics and Computational Biology (BiRBi). Faculty of Mathematics, University of Belgrade, Studentski trg 16, 11000 Belgrade.
- Member, The Data Visualization Society (DVS), EIN 84-1923759. 241 G St NW, Washington, DC 20001, USA.

Driving Licence: Category B (cars)

Interests: Linux, FOSS, Science Fiction, The Matrix, Video games, Hiking, Surfing, Drawing